



Chapter 2

Waves and Signals



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2.2 Waves

Wave is defined as a disturbance or vibration that travels through a medium or vacuum from one to another location. This disturbance carries energy from one location to another. We can also describe wave as the transfer of energy from one location to another.

Based on their mode of travel, waves are categorized into:

1. Electromagnetic Waves

An Electromagnetic wave can travel through vacuum and does not require material medium for transmission.

2. Mechanical Waves

It can not travel through vacuum and needs a material medium for transmission. Modern communication systems use electromagnetic waves to transfer information.



2.2.1 Wave Properties

Waves have properties that affect the information carried by them.

The various properties of a wave are -

1. Peak Amplitude.
2. Period.
3. Frequency.
4. Wavelength.
5. Phase.
6. Speed of propagation.



Peak Amplitude

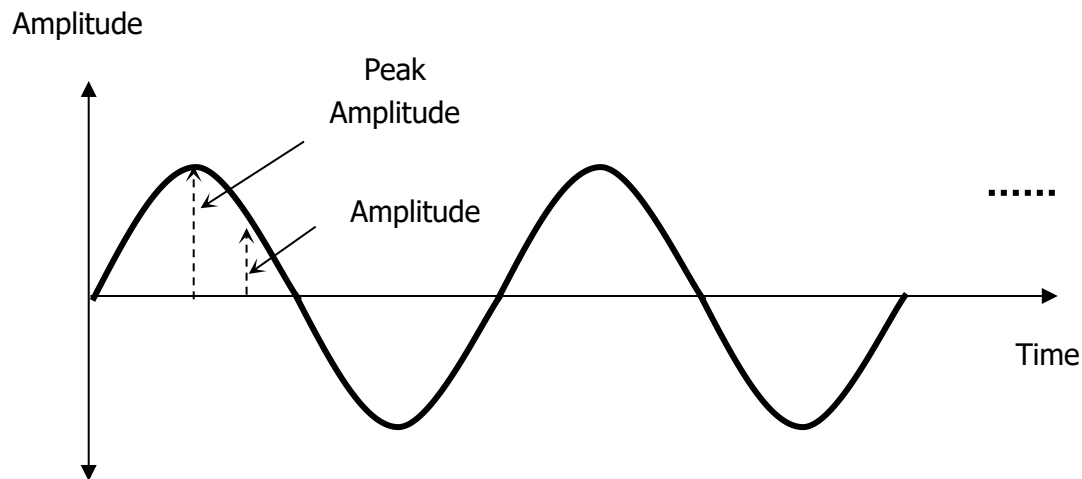


Figure 2.1: Peak Amplitude



Period and Frequency

Period refers to the amount of time, in seconds, a wave needs to complete one cycle. The period of a wave is measured in seconds. The frequency of a wave refers to the number of periods in one second. Its unit is Hertz (Hz).

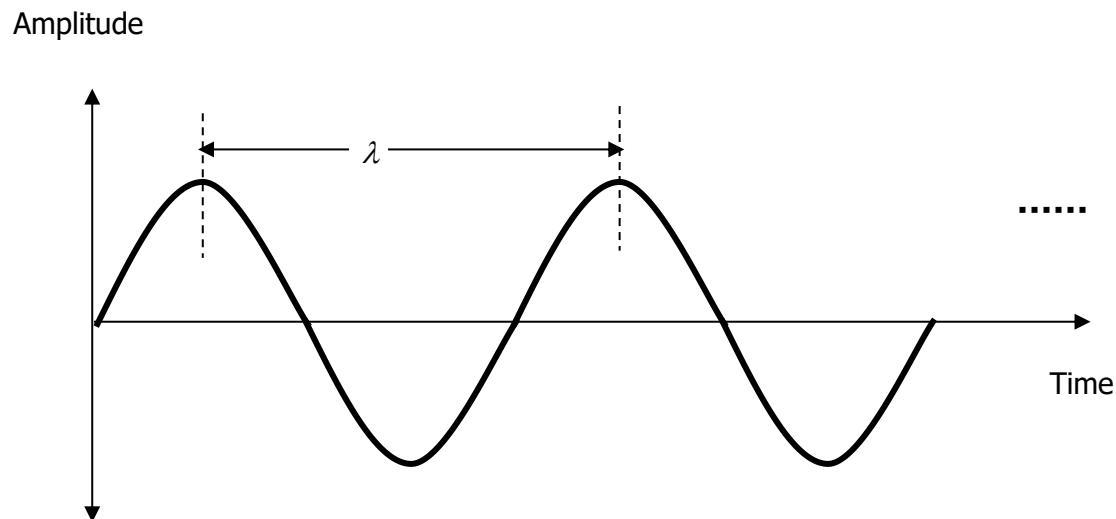


Figure 2.2: Period frequency and wavelength (λ) of a wave



Note

Frequency and period inverses each other i.e. $f = \frac{1}{T}$

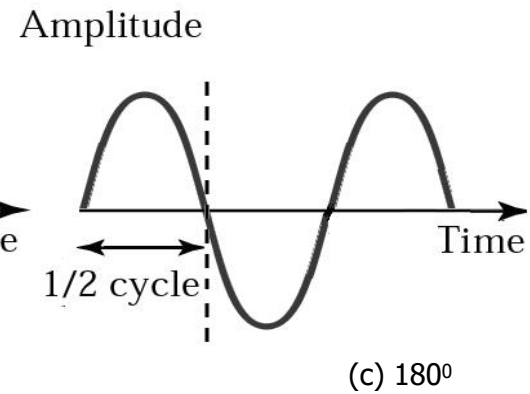
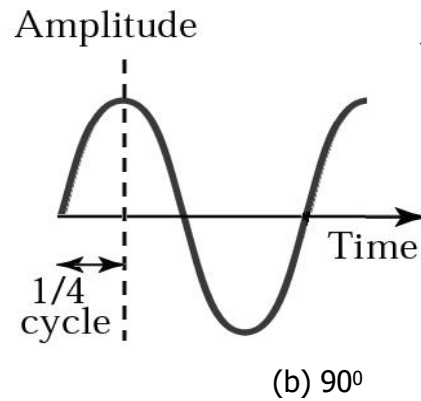
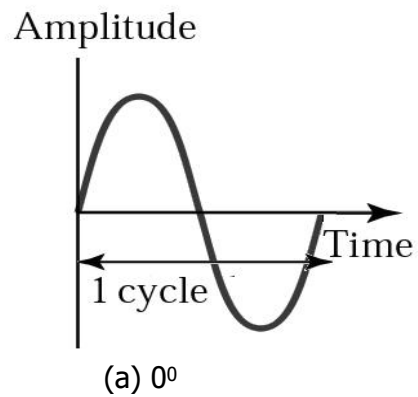


Note

The wavelength of a wave (λ) and the frequency of that wave is inversely related to each other $\lambda \propto \frac{1}{f}$



Phase





Note

The position of the waveform relative to time zero is called Phase.



Example 2.1

A sine wave is offset one-sixth of a cycle with respect to time zero. What is its phase in degrees and radians?

Solution

We know that one complete cycle is 360 degrees.

Therefore, 1/6 cycle is $(1/6) \times 360 = 60$ degrees $= 60 \times \frac{2\pi}{360}$ rad. $= 1.047$ rad.



Note

Electrical energy or electromagnetic energy is a form of energy present in any electric field or magnetic field, or in any volume containing electromagnetic radiation.



2.2.3 Propagation of EM Waves

The two types of electromagnetic waves that are used to carry information are-

1. Radio waves
2. Light waves

Based on the characteristics of propagation, radio waves are divided into three categories -

1. Ground waves
2. Sky waves
3. Line-of-sight waves.



2.2.4 Transmission Impairments

The four types of impairments are:

1. Attenuation.
2. Distortion.
3. Free space loss.
4. Noise.



1. Attenuation

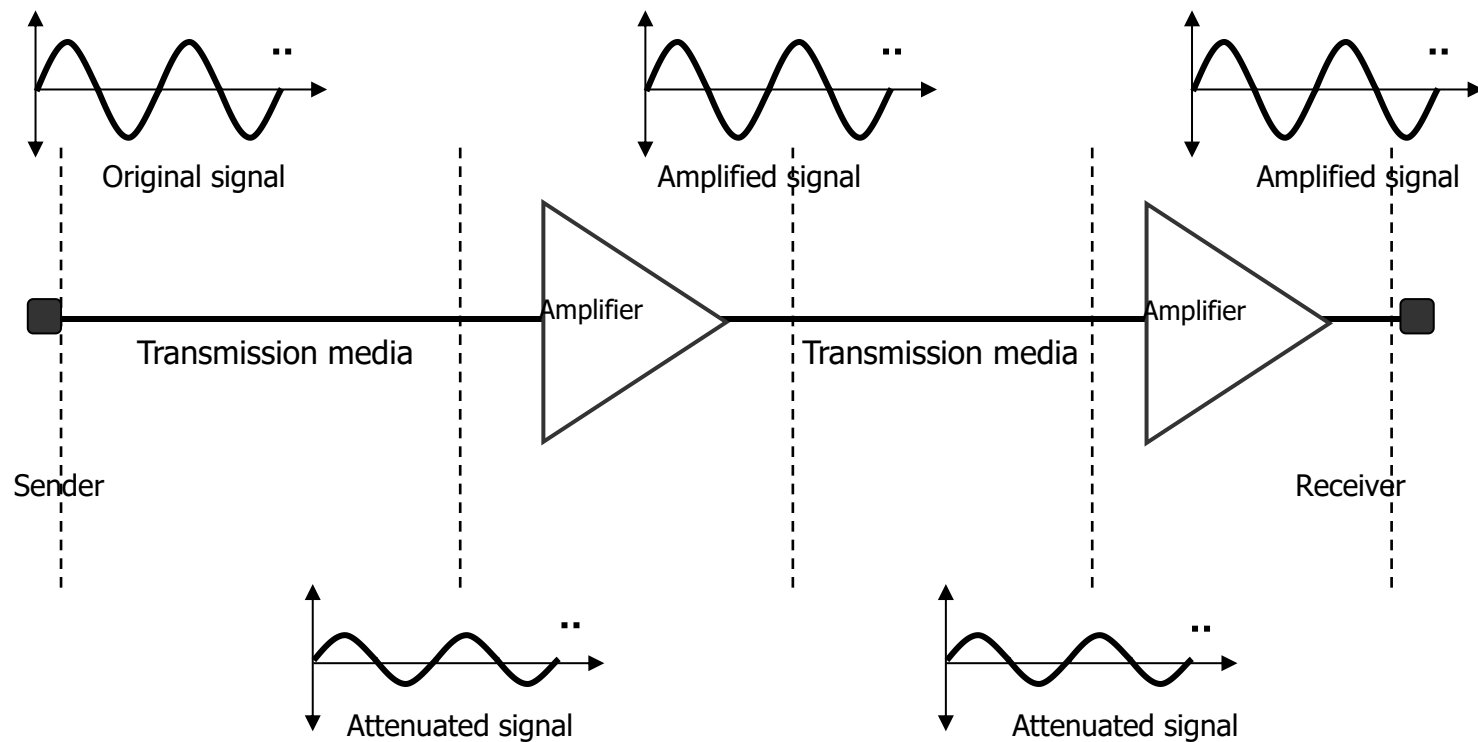


Figure 2.4: Attenuation



Example 2.2

A wave is transmitted through a medium. The power of the signal at the receiver is one-fourth of the power at the transmitter. Compute the attenuation.

Solution

According to the condition, the received power of the wave $P_r = 0.25P_t$

Using equation (2.4),

$$\begin{aligned}\text{Attenuation} &= 10 \log_{10} \left(\frac{P_r}{P_t} \right) \\ &= 10 \log_{10} \left(\frac{0.25P_t}{P_t} \right) \\ &= 10 \log_{10} (0.25) \\ &= -6.021 \text{ dB}\end{aligned}$$

Thus the attenuation is -6.021 dB



2. Distortion

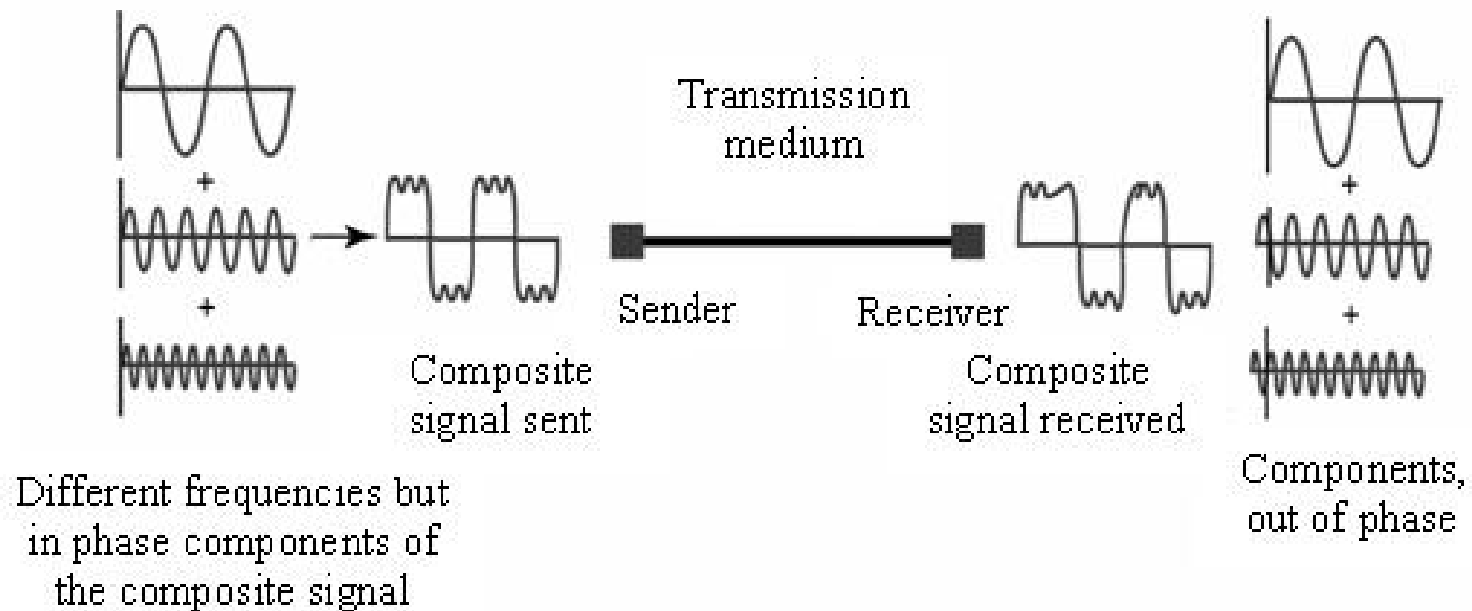


Figure 2.5: Distortion



3. Noise

Unwanted waves modify the waves carrying information during transmission. These unwanted waves are referred to as noise. Noise is divided into the following categories -

1. Thermal noise
2. Induced noise
3. Cross talk
4. Impulse noise



Note

The random motion of electrons in a medium produces thermal noise; thermal noise is also known as white noise or Johnson noise.



Example 2.3

A wave with frequency of 4000 Hz travels through space between two antennas. The distance between the two antennas is 50 meters. The wave is transmitted with a power of 250 W. Compute the power at the receiver and finds the free space loss for the wave.

Solution

$$\left(\frac{P_t}{P_r}\right) = \frac{(4\pi fd)^2}{c^2}$$

$$P_r = (P_t \times c^2) / (4\pi fd)^2 = (250 \times 3 \times 10^8) / (4\pi \times 4000 \times 50)^2 = 0.01186 \text{ W}$$

$$\text{Free Space Loss in dB} = 10 \log_{10} \left(\frac{P_t}{P_r} \right) = 10 \log_{10} (0.01186/250) = -43 \text{ dB}$$



Note

Data are defined as entities that convey meaning or information and signals are electric or electromagnetic representation of data.



2.3.1 Types of Signals

The three types of signals used in telecommunications are

1. Electrical signals
2. Light signals
3. Electromagnetic signals other than light (or radio waves)



2.3.2 Properties of Electrical Signals

- ✚ **Bandwidth of an Electrical Signal**
- ✚ **Data Rate of an Electrical Signal**



Note

The absolute value of the difference between the lowest and highest frequencies of a signal is called Bandwidth and the range of that a signal spans from minimum to maximum is called Spectrum.



2.3.3 Properties of Light Signals

- ✚ **Refraction**
- ✚ **Bandwidth**
- ✚ **Loss**
- ✚ **Data Rate**



Refraction

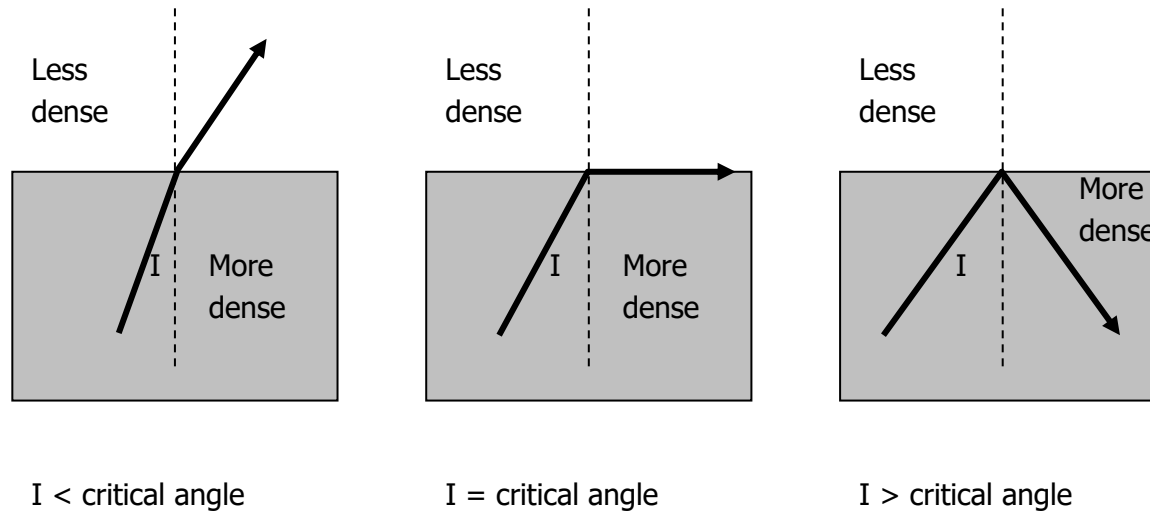


Figure 2.6: Bending of light ray



2.3.4 Forms of Data and Signals

The two types of signals used in a telecommunication network are-

1. Analog
2. Digital



Note

Signal is the electrical representation of data. Signal and data both have analog and digital forms.

Analog signals take an infinite number of values in a range. Digital signals take a limited number of values in a range.



Analog and Digital Signal

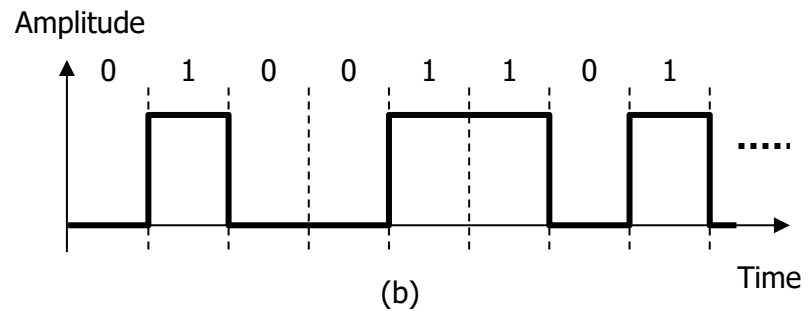
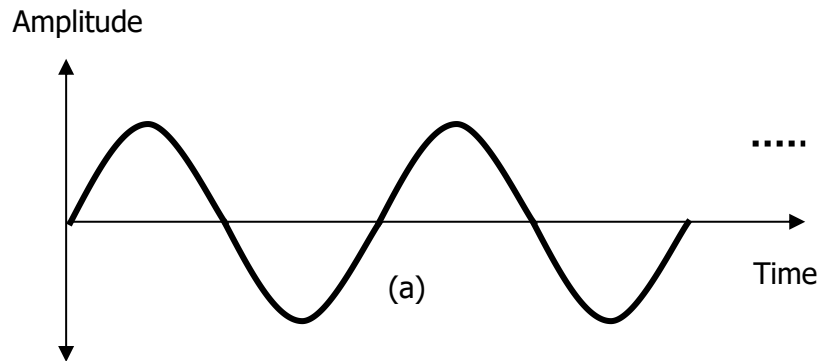


Figure 2.7: (a) Analog signal (b) Digital signal



Types of Analog Signal

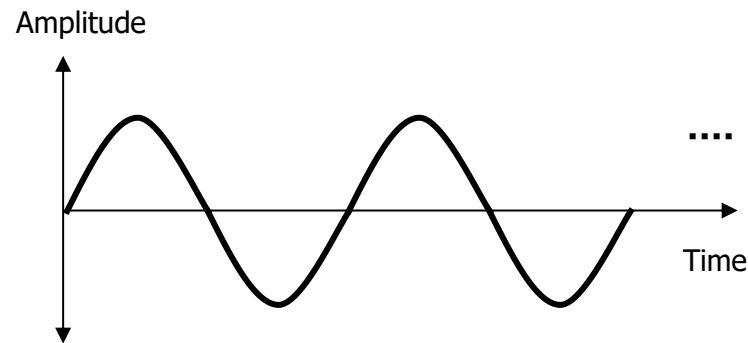


Figure 2.8: A simple analog signal

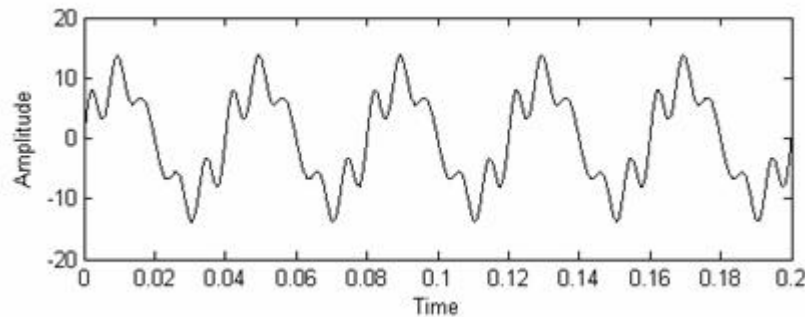


Figure 2.9: A simple analog signal



Bit Interval

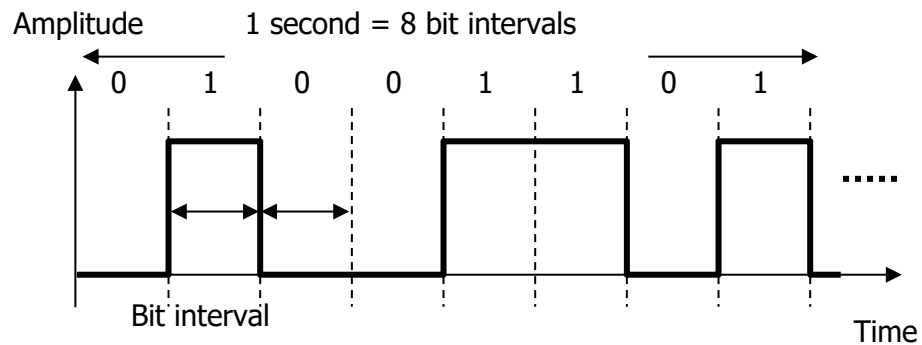


Figure 2.11: Bit interval and bit rate



Example 2.6

A digital signal produced in 5 seconds represents 300 bits. Find the bit interval for the signal.

Solution

A signal represents 300 bits in 5 seconds. Therefore, the time to represent one bit is $5/300$ seconds, that is, 0.017 seconds.

Example 2.7

The bit duration of a digital signal is 0.04 seconds. Find the bit rate of the signal.

Solution

The time duration of one bit is 0.04 seconds. Thus, the bit rate will be $1/0.04$ that is, 25 bps.



2.4 Data Rate Limits

Data rate depends on three factors

1. Available bandwidth.
2. Level of signals.
3. Quality of the channel.



Note

Signal levels refer to the number of values allowed in a particular signal and data levels refer to the number of values used to represent data.



Example 2.8

Determine bit rate for a noiseless channel with a bandwidth of 1000 Hz transmitting a signal with two signal levels.

Solution

Using equation (2.9), the maximum bit rate can be calculated as,

$$\text{Bit Rate} = 2 \times 1000 \times \log_2 2 = 2000 \text{ bps.}$$

Example 2.9

Let an extremely noisy channel in which the value of the signal-to-noise ratio is almost zero i.e. the noise is very strong that the signal is faint. Find the channel capacity

Solution

The capacity of this is calculated as [using equation (2.10)],

$$C = B \log_2 (1 + \text{SNR}) = B \log_2 (1 + 0) = B \log_2 (1) = B \times 0 = 0$$



2.5 Comparing Analog and Digital Signals

- 1. Impact of Noise**
- 2. Information Loss**
- 3. Introduction of Error**